complex case. In the remaining two cases, let B be the invariant bilinear form on V and let $(\ ,\ )$ be the invariant positive Hermitian form (they are defined up to a nonzero complex scalar and a positive real scalar, respectively). Define the operator $j : V \to V$ such that $B(v, w) = (v, jw)$. Show that j is complex antilinear ($ji = -ij$), and $j^2 = \lambda \cdot \mathrm{Id}$, where λ is a real number, positive in the real case and negative in the quaternionic case (if B is renormalized, j multiplies by a nonzero complex number z and j^2 by $z\bar{z}$, as j is antilinear). Thus j can be normalized so that $j^2 = 1$ in the real case and $j^2 = -1$ in the quaternionic case. Deduce the claim from this.

(b) Show that V is of real type if and only if V is the complexification of a representation $V_{\mathbb{R}}$ over the field of real numbers.

Example 5.1.3. For $\mathbb{Z}/n\mathbb{Z}$ all irreducible representations are of complex type except the trivial one and, if n is even, the "sign" representation, $m \to (-1)^m$, which are of real type. For S_3 all three irreducible representations $\mathbb{C}_+, \mathbb{C}_-, \mathbb{C}^2$ are of real type. For S_4 there are five irreducible representations $\mathbb{C}_+$, $\mathbb{C}_-$, $\mathbb{C}^2$, $\mathbb{C}^3_+$, $\mathbb{C}^3_-$, which are all of real type. Similarly, all five irreducible representations of A_5 — $\mathbb{C}$, $\mathbb{C}^3_+$, $\mathbb{C}^3_-$, $\mathbb{C}^4$, $\mathbb{C}^5$ — are of real type. As for Q_8, its 1-dimensional representations are of real type, and the 2-dimensional one is of quaternionic type.

Definition 5.1.4. The **Frobenius-Schur indicator** $FS(V)$ of an irreducible representation V is 0 if it is of complex type, 1 if it is of real type, and -1 if it is of quaternionic type.

Theorem 5.1.5 (Frobenius-Schur)**.** *The number of involutions (= elements of order ≤ 2) in G is equal to $\sum_V \dim(V) FS(V)$, i.e., the sum of dimensions of all representations of G of real type minus the sum of dimensions of its representations of quaternionic type.*

Proof. Let $A : V \to V$ have eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_n$. We have

$$\mathrm{Tr}\,|_{S^2V}(A \otimes A) = \sum_{i \leq j} \lambda_i \lambda_j,$$

$$\mathrm{Tr}\,|_{\Lambda^2 V}(A \otimes A) = \sum_{i < j} \lambda_i \lambda_j.$$